



FOOD DONATION & NGO CONNECT PLATFORM

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Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- Connects food donors with registered NGOs for efficient surplus collection. Reduces food waste and hunger using DBMS principles.
- Outcome:** Seamless distribution transparency.

Problem Statement & Objectives

- Problem:** Lack of real-time coordination leading to avoidable food waste.
- Obj 1:** Intuitive surplus food registration.
- Obj 2:** NGO tools to accept & manage pickups.
- Obj 3:** Secure MySQL relational backend with CRUD.
- Scope:** Donors, NGOs, Donations, Admin.

Existing vs Proposed System

Feature	Existing System	Proposed System
[Feature 1]	Pone calls	Digital Form
[Feature 2]	Delays	Real-Time dashboard
[Feature 3]	Spreadsheets	MYSQL ACID DB
[Feature 4]	Low visibility	Lifecycle Traking

Feasibility Study

- Technical:** HTML/CSS/JS, Spring Boot, MySQL.
- Economic:** Open-source stack, zero licensing.
- Operational:** Low-friction UI for users.
- Schedule:** Completion within semester timeline.

Results & Testing

- Verified all core CRUD operations & workflows.
- REST APIs tested & validated via Postman.

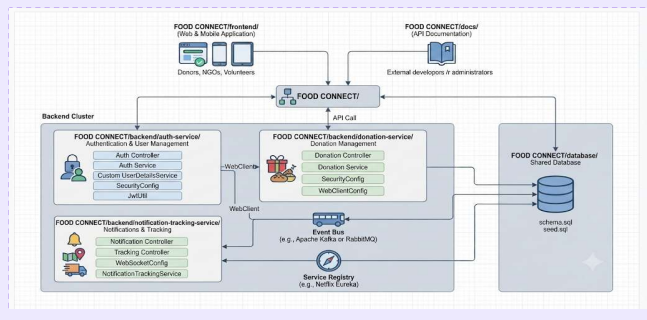
Conclusion

- Developed the **Food Connect** microservices platform using Spring Boot and React.js to streamline surplus food redistribution.

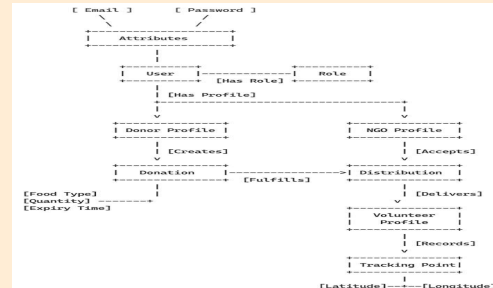
Tools & Technologies

- Front-End: HTML5, CSS3, JavaScript
- Back-End: Java / Spring Boot + REST APIs
- Database: MySQL Server via JDBC
- Tools: Git & GitHub, Postman, VS Code, Draw.io

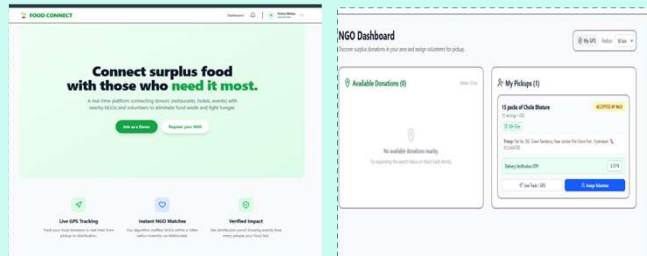
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Integrate automated AI-driven route optimization and matching algorithms to pair surplus food donors with the closest active NGOs and volunteers in real time.

